

Problem 4. Let $W \subset \mathbb{R}^n$ be a linear subspace. In class, we defined $W^\perp$ as the set of vectors

$$\{v \in \mathbb{R}^n \mid v \cdot w = 0 \text{ for all } w \in W\}.$$

- (a) Prove from the definition that $W^\perp$ is a linear subspace of $\mathbb{R}^n$.
- (b) Prove that $W \cap W^\perp = \{0\}$, by showing that if $w \in W$ and $w \in W^\perp$, then $w = 0$.
- (c) Suppose that $v_1 + v_2 = v'_1 + v'_2$, where $v_1, v'_1 \in W$ and $v_2, v'_2 \in W^\perp$. Show that $v_1 = v'_1$ and $v_2 = v'_2$. That is, if a vector v can be decomposed as $v_1 + v_2$ for $v_1 \in W$, $v_2 \in W^\perp$, then this decomposition is unique.

Solution. Let $W \subset \mathbb{R}^n$ be a linear subspace, and

$$W^\perp = \{v \in \mathbb{R}^n \mid v \cdot w = 0 \text{ for all } w \in W\}$$

- (a) We need to check the following conditions:

- (1) $0 \in W^\perp$
- (2) If $v_1, v_2 \in W^\perp$, then $v_1 + v_2 \in W^\perp$
- (3) If $v \in W^\perp$, $c \in \mathbb{R}$, then $cv \in W^\perp$.

For (1), $0 \cdot w = 0$ for any vector w , in particular for any $w \in W$, so $0 \in W^\perp$. For (2), if $v_1, v_2 \in W^\perp$, then for any $w \in W$,

$$(v_1 + v_2) \cdot w = v_1 \cdot w + v_2 \cdot w = 0 + 0 = 0$$

and so $v_1 + v_2 \in W^\perp$. For (3), if $v \in W^\perp$ then for any $w \in W$,

$$(cv) \cdot w = c(v \cdot w) = c(0) = 0$$

and so $cv \in W^\perp$. We conclude that $W^\perp$ is a linear subspace.

- (b) Suppose that $w \in W$ and $w \in W^\perp$. Then in particular $w \cdot w = 0$. If

$$w = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

then $0 = w \cdot w = x_1^2 + \cdots + x_n^2$. As for each i , $x_i^2 \geq 0$, the only way for this to be 0 is if each $x_i = 0$, so $w = 0$.

- (c) Suppose that $v_1 + v_2 = v'_1 + v'_2$, for $v_1, v'_1 \in W$, $v_2, v'_2 \in W^\perp$. We can rearrange this equation as

$$v_1 - v'_1 = v'_2 - v_2$$

Call this vector u . Then as $u = v_1 - v'_1$, $u \in W$, and as $u = v'_2 - v_2$, $u \in W^\perp$, using the fact that W and $W^\perp$ are linear subspaces. From part (b), this implies that $u = 0$, and so $v_1 = v'_1$, $v_2 = v'_2$.